

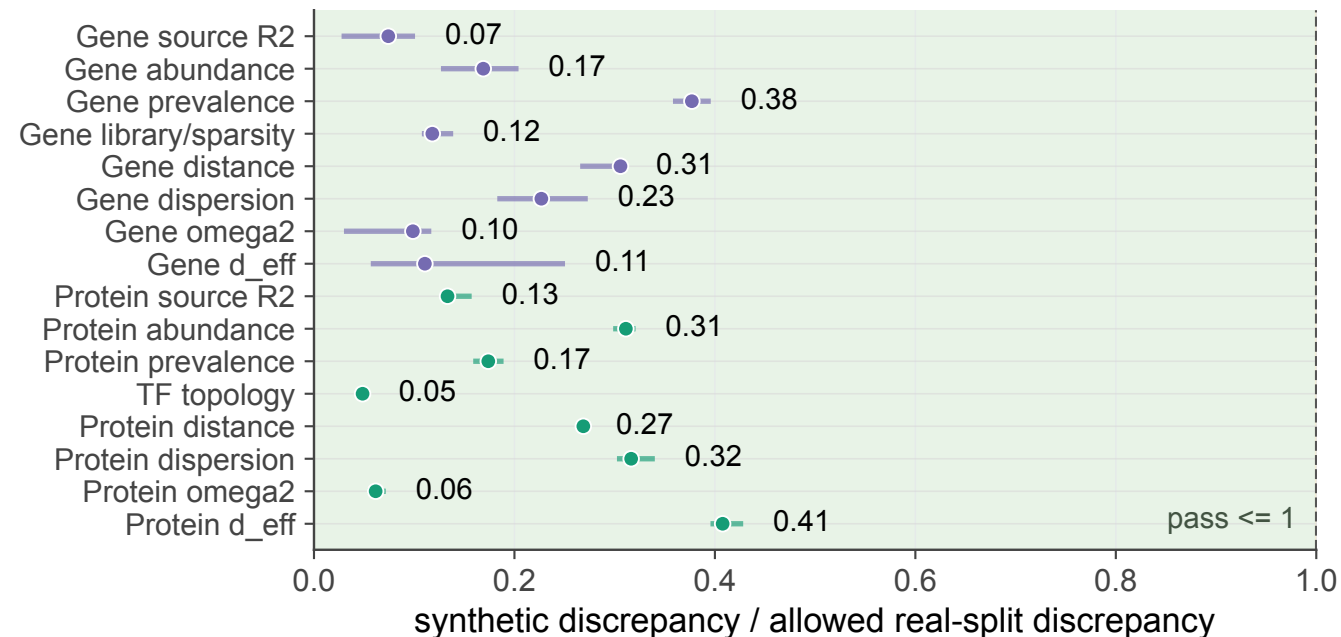
a Feature-level fidelity

Gene abundance fidelity

Protein abundance fidelity

b

Real-split calibrated fidelity



c Power-relevant fidelity

Gene distance

KS w/b 0.15/0.11
low-tail 7.6%
zero 0.5%
 ω^2 FE: 1.11 [1.03, 1.13]
d_eff FE: 1.05 [1.03, 1.12]

— real within
- - synthetic within
— real between
- - synthetic between

Protein distance

KS w/b 0.08/0.12
low-tail 8.1%
zero 0.0%
 ω^2 FE: 1.07 [1.06, 1.09]
d_eff FE: 1.18 [1.17, 1.19]

d Protein taxon-function structure

node degree

KS taxon/function
0.001/0.003

node prevalence

rank r taxon/function
0.998/0.999

edge support

edge KS 0.017
connectance ratio 0.974

synthetic quantile

real quantile

synthetic prevalence

real prevalence

synthetic quantile

real quantile

taxon degree
function degree

taxa
functions